UNIVERSITY OF THE PEOPLE

MATH 1280-01 - AY2026-T4

Assignment Activity Unit 7

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Continuous Probability Distributions

1. Continuous Probability Density Function (CPDF)

A continuous probability density function (PDF) describes the probability distribution of a continuous random variable, where values can take any number within an interval. Unlike discrete distributions (which deal with countable outcomes like number of passengers), a continuous distribution assigns probability through area under a curve, not direct values.

In mathematical terms, for a continuous random variable *XX*X, probabilities are calculated as:

$$P(a < X < b) = \int_a^b f(x)\,dx$$

where *f(x)f(x)*f(x) is the probability density function.

A key difference from a discrete probability distribution is that in discrete cases, we calculate probabilities at exact values (like $P(X=x)$, while in continuous distributions:

$P(X=c)=0$

because a single point has no width (no area).

Airport Example

In the airport scenario, check-in times between 5 and 15 minutes form a continuous distribution. Instead of counting passengers with exact times, we measure the probability of time intervals like 7–10 minutes using area under the curve.

2. Uniform Distribution Probability (Check-in Time)

Given:

- Uniform distribution between $a=5$ and $b=15$
- Find $P(7<X<10)$

For a uniform distribution:

$$f(x) = \frac{1}{b-a}$$

$$f(x) = \frac{1}{15-5} = \frac{1}{10}$$

Now calculate probability:

$$P(7 < X < 10) = (10-7) \times \frac{1}{10}$$

$$= 3 \times \frac{1}{10} = \frac{3}{10} = 0.3$$

Final Answer:

P(7<X<10) = 0.3 (30%)

3. Exponential Distribution Probability (Late Arrivals)

Given:

- Mean $\mu=10$
- Rate parameter:

$$\lambda = \frac{1}{\mu} = \frac{1}{10} = 0.1$$

We are asked:

P(X<5)

For exponential distribution:

$$P(X < x) = 1 - e^{-\lambda x}$$

$$P(X < 5) = 1 - e^{-0.1 \cdot 5}$$
$$= 1 - e^{-0.5}$$

$$e^{-0.5} \approx 0.6065$$
$$P(X < 5) = 1 - 0.6065 = 0.3935$$

$$P(X < 5) \approx 0.3935 \quad (39.35\%)$$

Conclusion

Continuous probability distributions are essential in real-world systems like airport operations because they model time-based events smoothly rather than counting fixed outcomes. The uniform distribution helps represent consistent processing times between fixed limits (5–15 minutes), while the exponential distribution is useful for modeling unpredictable waiting times such as late passenger processing. These tools allow airport management to predict delays, optimize passenger flow, and improve operational efficiency using probability-based decision making.

References

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